Customer Segmentation & Service Recommendation System

Major Project Report — Data Science Track | Veda Technology

Abstract

An educational customer analytics workflow using synthetic data to group customers by behavior, engagement, spending, inquiries, support requests, and training interest. K-Means discovers segments; PCA visualizes them; transparent rules suggest relevant services.

Problem Statement

Customers have different interaction patterns and needs. This project demonstrates how behavioral data can be organized into interpretable segments and linked to service suggestions.

Objectives

Prepare data, handle missing values, standardize features, apply K-Means, visualize clusters with PCA, assess silhouette score, summarize segment profiles, and deliver a simple dashboard.

Dataset

300 artificial records with customer_id, website_visits, engagement_score, monthly_spend, service_inquiries, support_requests, training_interest, and preferred_channel. No confidential company data is used.

Tools

Python, Pandas, NumPy, Scikit-learn, Matplotlib, Streamlit, CSV, and GitHub.

Methodology

Numeric fields are cleaned and missing values replaced with feature medians. StandardScaler normalizes features. K-Means assigns clusters for configurable K. PCA projects the standardized data into two dimensions. Segment means support interpretation.

Evaluation

The dashboard reports the silhouette score, a diagnostic of cluster cohesion and separation. It is not proof of business value. Compare K values and inspect profiles before naming segments.

Recommendation Logic

High-spend, high-engagement customers receive premium-service suggestions; training-oriented customers receive learning suggestions; customers with frequent inquiries/support requests receive consultation/support suggestions; lower-activity users receive introductory resources. Rules are illustrative.

Dashboard

The Streamlit app shows customer and segment counts, silhouette score, segment distribution, PCA scatter plot, segment averages, recommendations, and downloadable results.

Limitations

Synthetic data may not represent real behavior. K-Means depends on feature choice, scaling, and K. PCA can hide higher-dimensional structure. Rule-based suggestions are not a validated recommender.

Future Scope

Test other clustering methods, compare K using elbow and silhouette methods, add privacy-safe real data with authorization, build a Power BI dashboard, and evaluate recommendations using feedback and measured outcomes.

Conclusion

This project demonstrates an end-to-end introductory workflow for customer segmentation and service recommendation, while distinguishing illustrative synthetic results from validated business insights.